Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2022 | ENGI3451-WE01 |

**Title: (BEng/MEng)**

Electronics and Communications 3

| | |
|---|---|
| Time Allowed: | 2 hours |
| Additional Material provided: | No |
| Reporting problems during the exam: | Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk.<br><br>If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk. |

| | |
|---|---|
| **Instructions to Candidates:** | Answer ALL questions.<br><br>Please answer each question on a new page.<br><br>All workings must be shown.<br><br>This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window.<br><br>• *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.*<br><br>• *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* |

| Revision: | |
|---|---|

# Section A

## Question 1

(a) Consider a carrier signal $c(t) = 2\cos(4 \times 10^5 \pi t)$ and the periodic message signal shown in Figure Q1. Assume double-sideband (DSB) amplitude modulation.

(i) Determine the modulation index.

(ii) Determine the power efficiency.

(iii) Discuss how to improve the power efficiency/consumption.

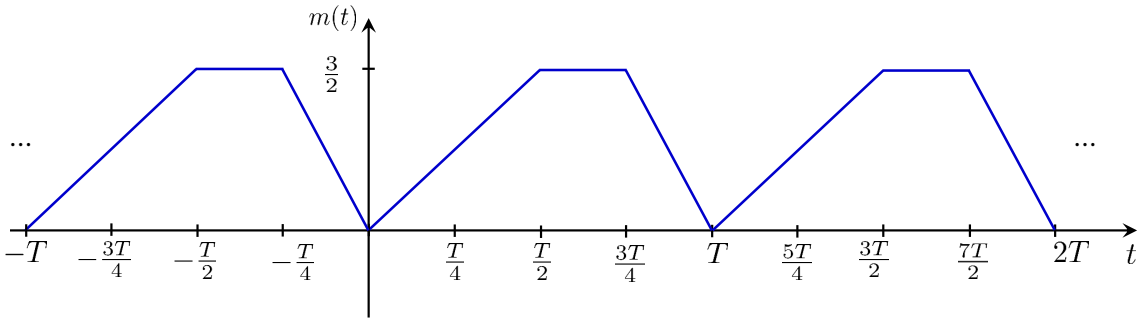



Figure Q1: A message signal.

**[45%]**

(b) Consider the analogue signal $x(t) = 2\cos(2000\pi t) + 3\cos(5000\pi t + \frac{\pi}{3})$ to be converted into a digital signal using an analogue-to-digital convertor.

(i) Determine the minimum sampling rate that should be used to guarantee a guard band of 0.5 kHz.

(ii) What would happen if the used sampling rate is 3.5 kHz?

**[25%]**

(c) The generator matrix of a (6, 3) linear code is

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

The code is extended by adding an overall parity check bit to each codeword so that the weight of each resulting codeword is even. What are the rate and minimum distance of the extended code?

**[30%]**

**continued**

## Question 2

(a) An angle-modulated signal is given by

$$s(t) = 10\cos\left(4\times 10^7\pi t + 2\sin(2000\pi t) + 3\cos(10000\pi t - \frac{\pi}{2})\right)$$

(i) Determine the frequency deviation.

(ii) Determine the modulation index.

(iii) Explain whether this is a frequency modulated (FM) or a phase modulated (PM) signal.

**[40%]**

(b) A discrete memoryless source produces outputs $a$, $b$, $c$, $d$ and $e$ with probabilities $p(a) = \frac{1}{3}$, $p(b) = \frac{1}{3}$, $p(c) = \frac{1}{9}$, $p(d) = \frac{1}{9}$ and $p(e) = \frac{1}{9}$, respectively. Assume coding over an alphabet of size $D = 3$ (ternary). Note that a ternary code is one which uses three alphabets in coding, e.g., $\{0, 1, 2\}$.

(i) Design a Huffman code for this source and determine its average codeword length.

(ii) Determine the entropy of this source. Comment on the result.

(iii) Now assume that instead of the true probabilities of the outputs above, you use the following probabilities: $q(a) = 0.3$, $q(b) = 0.3$, $q(c) = 0.2$, $q(d) = 0.1$ and $q(e) = 0.1$ for encoding. Encode the above source using Huffman coding using the probability mass function $q$ and determine the average codeword length. Explain the result.

**[50%]**

(c) Consider two independent discrete random variables $X$ and $Y$, taking values in the alphabets $\mathcal{X}$ and $\mathcal{Y}$, with marginal probability mass functions (PMFs) $p_X(x)$ and $p_Y(y)$, respectively, and joint PMF $p_{XY}(x, y)$. Find the mutual information $I(X;Y)$ and explain the result.

**[10%]**

**continued**

# Section B

## Question 3

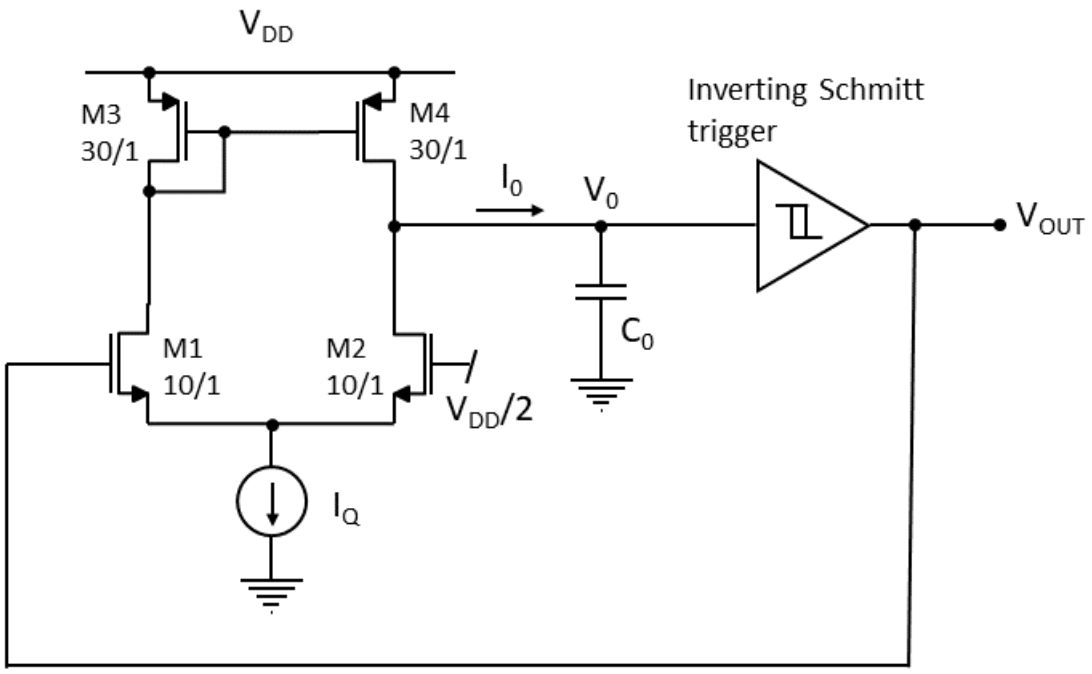



Figure Q3: A relaxation oscillator.

Figure Q3 shows the circuit of a relaxation oscillator. The capacitor $C_0$ at the input of the inverting Schmitt trigger is charged by the output current of the differential amplifier $I_0$.
Assume that the MOSFETs have the following parameters:

- $\mu_n C_{OX} = 180\ \mu\text{A/V}^2$.
- $\mu_p C_{OX} = 60\ \mu\text{A/V}^2$.
- $|V_T| = 0.5$ V for all MOSFETs.

The aspect ratios W/L of the MOSFETs' channels are shown in the circuit. Assume that any channel modulation effect is negligible. Additionally, $V_{DD} = 5$ V, $I_Q = 60\ \mu$A and $C_0 = 1$ nF.
The output of the Schmitt trigger goes to 0 V when $V_0 > 3$ V and goes to $V_{DD}$ when $V_0 < 2$ V.

(a) Explain, in your own words, how this circuit operates and derive an expression for the oscillation frequency. All steps must be shown.

**[50%]**

(b) Design a circuit which implements the Schmitt trigger. Assume that an ideal comparator is available.

[**30%**]

(c) Propose a change in the circuit of this oscillator to enable the control of the frequency using an external voltage.

[**20%**]

## Question 4

(a) Explain, in your own words, what are the key advantages of the CMOS inverter when compared to a simple common-source amplifier. Use graphs and diagrams to support your answer.

**[20%]**

(b) Figure Q4 shows the diagram of a simple state machine with two binary inputs, A and B, and a binary output OUT. There is only one state which is stored in the D flip-flop.

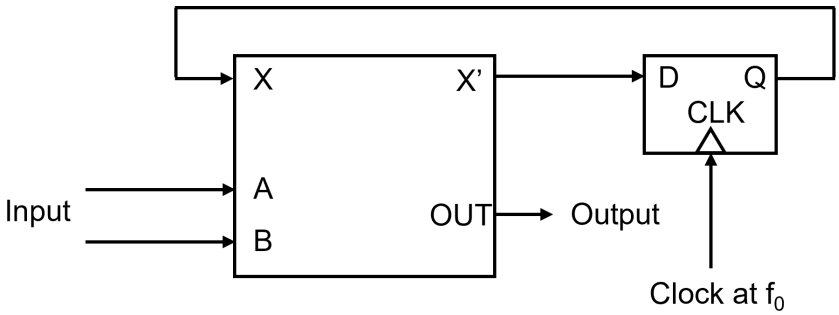



Figure Q4: The state machine considered in part (b).

The truth table for OUT and the next state X' is:

| X | A | B | X' | OUT |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 1 |

(i) Draw a CMOS circuit which implements this state machine. You will need to consider the combinatory logic circuits for X' and OUT as well as the D flip flop.

(ii) Assume that a prototype inverter has already been designed; the N-MOSFET has an aspect ratio $(W_N/L_N)$ = (64 nm/32 nm) and the P-MOSFET has an aspect ratio $(W_P/L_P)$ = (256 nm/32 nm). Dimension all MOSFETs in the combinatory logic circuits (do not dimension the MOSFETs in the flip-flop) in order to achieve a symmetric operation and a driving capacity as the prototype inverter in the worst case.

**[60%]**

**continued**

(c) Consider the combinatory logic circuits that you have designed in part (b). Assume that $V_{DD} = 3$ V, there is a load capacitance $C_L = 1$ fF on both OUT and X', and that $C_{OX} = 150$ nf/cm$^2$. Find the dynamic power dissipation when X $= 0$, A $= 1$, and B is clocked at a frequency $f_0 = 1$ GHz. Only account for the MOSFET gate capacitances ($C_G = W \times L \times C_{OX}$) and the load capacitance $C_L$.

**[20%]**

**END**